6.4 How can electronic systems offer functionality to design solutions?

a. Demonstrate an understanding of how electronic systems provide input, control and output process functions, including:

   i. switches and sensors, to produce signals in response to a variety of inputs
   ii. programmable control devices
   iii. signal amplification
   iv. devices to produce a variety of outputs including light, sound, motion.

Considerations	*Maths & Science*
b. Demonstrate an understanding of the function of an overall system, referring to aspects including: i. passive components: resistors, capacitors, diodes ii. inputs: sensors for position, light, temperature, sound, infra-red, force, rotation and angle iii. process control: programmable microcontroller iv. signal amplification: MOSFET, driver ICs v. outputs: LED, sounder, solenoid, DC motor, servo motor, stepper motor, piezo actuator, displays vi. analogue and digital signals and conversion between them vii. open and closed loop systems including feedback in a system and how it affects the overall performance viii. sub-systems and systems thinking.	
c. Demonstrate an understanding of what can be gained from interfacing electronic circuits with mechanical and pneumatic systems and components, such as: • the ability to add electronic control as an input to mechanical or pneumatic output • the use of flow restrictors to control cylinder speed • the use of sensors to measure rotational speed, strain/force, distance.	
d. Demonstrate an understanding of networking and of communication protocols, including: i. wireless devices, such as: RFID, NFC, Wi-Fi, bluetooth ii. embedded devices iii. smart objects iv. networking electronic products to exchange information.	
e. Demonstrate an understanding of the basic principles of electricity, including: i. voltage ii. current iii. Ohm's law iv. power.	